

Yuki Ohnishi

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Academic Employment

University of Florida	Gainesville, FL
– Assistant Professor, Department of Statistics	Aug 2026 - Present
Yale School of Public Health	New Haven, CT
– Postdoctoral Associate, Department of Biostatistics	Aug 2024 - Jul 2026
Purdue University	West Lafayette, IN
– Graduate Teaching/Research Assistant, Department of Statistics	Aug 2018 - May 2024

Education

Purdue University	West Lafayette, IN, USA
PhD in Statistics	May 2024
– Dissertation: <i>Causal Inference in the Face of Assumption Violations</i>	
– Advisors: Jordan Awan and Arman Sabbaghi	

Research Interests

Methodology: Causal inference, Causal mediation analysis, Post-treatment confounders, Unobserved confounders, Heterogeneous treatment effects, Subgroup identification, Sequential treatment, Semi-competing risks, Interference, Bayesian nonparametrics, Differential privacy

Application: Clinical trials, Cluster randomized trials, Digital marketing

Publications

Referred Publications

- Ohnishi, Y. and F. Li. (2025) “A Bayesian nonparametric approach to mediation and spillover effects with multiple mediators in cluster-randomized trials,” *Journal of the American Statistical Association*, accepted.
- Ohnishi, Y., B. Karmakar and A. Sabbaghi. (2025). “Degree of Interference: A General Framework for Causal Inference under Interference,” *Journal of Machine Learning Research* 26 (120): 1-37.
- Ohnishi, Y., B. Karmakar and W. Kar. (2025). “Inferring Causal Effect of a Digital Communication Strategy under a Latent Sequential Ignorability Assumption and Treatment Noncompliance,” *Journal of the American Statistical Association* 120 (550): 685–97.
- Ohnishi, Y. and J. Awan. (2025). “Locally Private Causal Inference for Randomized Experiments,” *Journal of Machine Learning Research* 26 (14): 1-40.
- Ohnishi, Y. and A. Sabbaghi. (2024). “A Bayesian Analysis of Two-Stage Randomized Experiments in the Presence of Interference, Treatment Nonadherence, and Missing Outcomes,” *Bayesian Analysis* 19 (1): 205-234.
- Ohnishi, Y. and J. Honorio. (2021). “Novel change of measure inequalities with applications to PAC-Bayesian bounds and Monte Carlo estimation,” *International Conference on Artificial Intelligence and Statistics*, 1711-1719.

Other Publications

- Ohnishi, Y. and S. Sugaya. (2019). “Applying Bayesian Hierarchical Probit Model to Interview Grade” Evaluation. KDD’19, International Workshop on Talent and Management Computing.
- Ohnishi, Y. and S. Sugaya. (2018). “Bayesian Hierarchical Bernoulli-Weibull Mixture Model for Extremely Rare Events.” INFORMS Conference on Business Analytics & Operations Research.

Submitted Papers & Preprints

- Ohnishi, Y. and F. Li. “The Resolution of Causal Heterogeneity.”

2. **Ohnishi, Y.** and F. Li. “Sensitivity analysis for causal mediation: bridge score, sharp sensitivity bounds, and calibration.”
3. **Ohnishi, Y.** and F. Li. “Bayesian change-plane regression.”
4. **Ohnishi, Y.**, M. J. Daniels, L. Yang and F. Li. “Identification and estimation of causal mechanisms in cluster-randomized trials with post-treatment confounding using Bayesian nonparametrics.”
5. Liu, R., J. L. Warren, **Y. Ohnishi**, D. Spiegelman, L. Hu, and F. Li. “Calibrated Bayes analysis of cluster-randomized trials.”
6. **Ohnishi, Y.**, M. O. Harhay, G. Tong and F. Li. “Principal stratification with recurrent events truncated by a terminal event: A nested Bayesian nonparametric approach.”
7. **Ohnishi, Y.** and J. Awan. “Differentially Private Covariate Balancing Causal Inference.”

Presentations

†: invited talk

†The 39th New England Statistics Symposium (NESS 2026) Storrs, CT
 Identification and estimation of causal mechanisms in cluster-randomized trials with post-treatment confounding using Bayesian nonparametrics. May 2026

University of Florida, Department of Biostatistics, Department Seminar Gainesville, FL
 A Bayesian nonparametric approach to mediation and spillover effects with multiple mediators in cluster-randomized trials. Jan 2026

Texas A&M University, Department of Statistics, Department Seminar College Station, TX
 A Bayesian nonparametric approach to mediation and spillover effects with multiple mediators in cluster-randomized trials. Jan 2026

University of Florida, Department of Statistics, Department Seminar Gainesville, FL
 A Bayesian nonparametric approach to mediation and spillover effects with multiple mediators in cluster-randomized trials. Dec 2025

2025 Joint Statistical Meetings (JSM 2025) Nashville, TN
 A Bayesian nonparametric approach to mediation and spillover effects with multiple mediators in cluster-randomized trials. August 2025

The 38th New England Statistics Symposium (NESS 2025) New Haven, CT
 A Bayesian nonparametric approach to mediation and spillover effects with multiple mediators in cluster-randomized trials. June 2025

Purdue University, Department of Statistics Student seminar West Lafayette, IN
 Causal inference in the face of assumption violations. April 2024

†The University of Tokyo Todai Bayes seminar Tokyo, Japan
 A Bayesian Analysis of Two-Stage Randomized Experiments in the Presence of Interference, Treatment Nonadherence, and Missing Outcomes. July 2024

Midwest Machine Learning Symposium (MMLS 2023) Chicago, IL
 Locally Private Causal Inference. May 2023

†International Conference on Design of Experiments (ICODOE 2023) Memphis, TN
 A Bayesian Analysis of Two-Stage Randomized Experiments in the Presence of Interference, Treatment Nonadherence, and Missing Outcomes. May 2023

INFORMS Workshop on Quality, Statistics, and Reliability Indianapolis, IN
 A Bayesian Analysis of Two-Stage Randomized Experiments in the Presence of Interference, Treatment Nonadherence, and Missing Outcomes. 2022

The 24th International Conference on Artificial Intelligence and Statistics (AISTAT 2021)
 Novel Change of Measure Inequalities with Applications to PAC-Bayesian Bounds and Monte Carlo Estimation. April 2021

The 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD 2019), Workshop on Talent and Management Computing. Anchorage, AK
 Applying Bayesian Hierarchical Probit Model to Interview Grade Evaluation. August 2019

INFORMS Business Analytics & Operations Research Baltimore, MD
 Bayesian Hierarchical Bernoulli-Weibull Mixture Model for Extremely Rare Events. April 2018

Teaching Experiences

Big Data Summer Immersion at Yale (BDSY) program, Instructor	Summer 2025
STAT 598 Differential Privacy, Teaching Assistant	Fall 2022
STAT 538 Probability Theory I (Measure theoretic probability), Teaching Assistant	Spring 2021
STAT 539 Probability Theory II (Stochastic processes), Teaching Assistant	Fall 2021
STAT 350 Introduction to Statistics, Teaching Assistant	Spring 2020 - Fall 2021
STAT 190 Data Mine, Teaching Assistant	Fall 2018 - Fall 2020
STAT 113 Statistics and Society, Teaching Assistant	Fall 2018 - Fall 2020

Honors and Awards

Yale University-Mayo Clinic CERSI Scholars Award, 2025-2026.

JSM Early Career Award, Statistics in Epidemiology Section, American Statistical Association, 2025.

Research Recognition GRAD Award, Department of Statistics, Purdue University, 2024.

College of Science Graduate Student Travel Award, Purdue University, 2023.

Research and teaching fellowship, Purdue University, 2018-2024

Academic Services

Referee

<i>Annals of Applied Statistics</i>	<i>Journal of Applied Statistics</i>
<i>Bayesian Analysis</i>	<i>Journal of Causal Inference</i>
<i>Biometrics</i>	<i>Journal of the American Statistical Association (TM/ACS)</i>
<i>Computational Statistics and Data Analysis</i>	<i>Journal of the Royal Statistical Society: Series C</i>
<i>International Journal of Biostatistics</i>	<i>Statistics in Medicine</i>

Other services

Joint Statistical Meetings, American Statistical Association, 2026

- Early Career Award, Statistics in Epidemiology Section, Reviewer

The 38th New England Statistics Symposium, 2025

- Student Paper Award Committee, Reviewer
- Session Chair

Community memberships

American Statistical Association (ASA), 2023–present

International Society for Bayesian Analysis (ISBA), 2023–present

New England Statistical Society (NESS), 2025-present

Industrial Careers

Boehringer Ingelheim <i>Data Scientist Summer Intern</i>	Ridgefield, CT, USA
BizReach, Inc. <i>Data Scientist</i>	Tokyo, Japan
Nomura Research Institute <i>Software Engineer</i>	Tokyo, Japan